

INFORMATION SCIENCE ATTACHMENT ASSIGNMENT

PROJECT: MACHINE LEARNING-BASED WEB-INTEGRATED CHATBOT FOR MFANO BORA AFRICA

ASSIGNMENT TITLE:

Information Management and Knowledge Base Development for a Website Chatbot

TASK 1: INFORMATION REQUIREMENTS ANALYSIS

Identify and document the information requirements of the chatbot and its intended users.

TASK 2: INFORMATION SOURCE IDENTIFICATION

Identify the sources of information that will be used to develop the chatbot's knowledge base.

TASK 3: INFORMATION COLLECTION

Collect relevant and verified information about Mfano Bora Africa for use by the chatbot.

TASK 4: INFORMATION CLASSIFICATION

Classify the collected information into appropriate information categories for chatbot retrieval.

TASK 5: KNOWLEDGE BASE DEVELOPMENT

Design and develop a structured knowledge base containing the verified information required by the chatbot.

TASK 6: INFORMATION ORGANIZATION

Develop an appropriate structure for organizing chatbot information to make retrieval efficient and accurate.

TASK 7: INFORMATION QUALITY ASSESSMENT

Evaluate the collected information for accuracy, completeness, relevance, consistency and currency.

TASK 8: INFORMATION CLEANING

Identify and remove duplicate, outdated, incomplete and inconsistent information from the collected data.

TASK 9: METADATA DEVELOPMENT

Develop appropriate metadata fields for organizing and describing the information stored in the chatbot knowledge base.

TASK 10: INFORMATION RETRIEVAL DESIGN

Design a method for retrieving relevant information from the knowledge base based on user questions.

TASK 11: FAQ DEVELOPMENT

Develop a structured collection of frequently asked questions and verified answers for integration into the chatbot knowledge base.

TASK 12: INFORMATION MAPPING

Map common user questions to the appropriate information categories, knowledge-base records and chatbot intents.

TASK 13: INFORMATION GAP ANALYSIS

Identify information gaps within the chatbot knowledge base and determine the information required to address them.

TASK 14: INFORMATION GOVERNANCE

Develop guidelines for maintaining, updating, verifying and approving information used by the chatbot.

TASK 15: INFORMATION SECURITY AND PRIVACY

Identify information security and privacy requirements that should be considered when storing and processing chatbot conversations and user information.

TASK 16: INFORMATION ACCESS AND RETRIEVAL TESTING

Test whether the required information can be retrieved accurately and efficiently from the knowledge base.

TASK 17: CHATBOT INFORMATION EVALUATION

Evaluate the relevance, accuracy and completeness of chatbot responses based on the information contained in the knowledge base.

TASK 18: KNOWLEDGE BASE MAINTENANCE PLAN

Develop a plan for regularly reviewing and updating the chatbot's knowledge base.

TASK 19: INFORMATION MANAGEMENT REPORT

Prepare a report documenting the information sources, collection process, classification, organization, quality assessment, knowledge base development and retrieval methodology.

TASK 20: RECOMMENDATIONS

Provide recommendations for improving the management, organization, retrieval, security and continuous updating of information used by the chatbot.